

# ProBuild Operations Platform (v2) — Test Execution Report

## Document control

| Field             | Value                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------|
| System under test | ProBuild Operations Platform — orchestration architecture (Camunda 8)                                                           |
| Processes         | Journey_Customer (3-pool collaboration), Cap_Sales , Cap_Warehouse , Cap_Loyalty , Cap_Hire , Partner_FinTrust , Partner_FixPro |
| Decision tables   | finance_assessment , loyalty_band , hire_pricing , maintenance_triage                                                           |
| Workers           | 24 @JobWorker handlers across seven capability packages                                                                         |
| Engine            | Camunda 8.8.11 Self-Managed (c8run) · Java 21 · Spring Boot 3.5.9                                                               |
| Tester            | MA                                                                                                                              |
| Execution date    | 13 July 2026                                                                                                                    |

## Approach

Each journey path is driven **through the live engine's REST API** end-to-end: an instance of `Journey_Customer` is started, user tasks are completed in sequence (across the Customer, FinTrust and FixPro pools), automatic service tasks and DMN business-rule tasks execute, and the outcome is asserted three ways — the **process reaches the expected end event**, the **called capabilities and partner instances complete**, and the **H2 database holds the correct rows**. The driver is `suite.mjs` ; raw results are in `docs/evidence/journey-results.json` .

## Pre-flight

| # | Check                                                  | Result |
|---|--------------------------------------------------------|--------|
| 1 | App starts; log shows all 7 processes + 4 DMN deployed | PASS   |
| 2 | 24 job workers registered                              | PASS   |
| 3 | Operate shows all processes at version, active         | PASS   |
| 4 | Seed: 3 trade cards, 11 inventory lines                | PASS   |

## Execution summary

| ID         | Scenario                         | Path exercised                                                                  | End event          | Result |
|------------|----------------------------------|---------------------------------------------------------------------------------|--------------------|--------|
| TC-CARD-01 | Apply Trade Card (new)           | orchestrator → Cap_Loyalty (issue + loyalty_band DMN)                           | Customer served    | PASS   |
| TC-CARD-02 | Renew Trade Card                 | orchestrator → Cap_Loyalty (renew + DMN)                                        | Customer served    | PASS   |
| TC-BUY-01  | Buy in stock, card payment       | Cap_Sales → Cap_Warehouse → Cap_Loyalty                                         | Customer served    | PASS   |
| TC-BUY-02  | Buy with finance <b>approved</b> | + FinTrust pool, finance_assessment DMN → APPROVE                               | Customer served    | PASS   |
| TC-BUY-03  | Buy with finance <b>declined</b> | FinTrust finance_assessment DMN → DECLINE (low score)                           | Customer cancelled | PASS   |
| TC-BUY-04  | Buy out of stock                 | Cap_Sales → backorder                                                           | Customer cancelled | PASS   |
| TC-BUY-05  | Buy, quote rejected              | Cap_Sales → quote → reject                                                      | Customer cancelled | PASS   |
| TC-HIRE-01 | Hire + <b>maintenance</b>        | Cap_Hire ( hire_pricing DMN) → Cap_Warehouse → FixPro ( maintenance_triage DMN) | Customer served    | PASS   |
| TC-HIRE-02 | Hire, clean return               | Cap_Hire → Cap_Warehouse (restock)                                              | Customer served    | PASS   |
| TC-HIRE-03 | Hire, tool unavailable           | Cap_Hire → unavailable                                                          | Customer cancelled | PASS   |

**10 / 10 journeys COMPLETED — 0 stuck, 0 incident.**

All four DMN decision tables were evaluated during the run (14 evaluation instances visible in Operate → Decisions: Finance credit assessment, Loyalty discount band, Tiered hire pricing, FixPro batch triage). See [docs/evidence/dmn-decisions-evaluated.png](#).

## Representative evidence (H2 assertions)

### TC-BUY-02 — buy with finance approved

ORDERS: ORD-... DISPATCHED, £129.90  
 FINANCE\_TRANSFERS: FIN-... APPROVED, £129.90, term 6 (finance\_assessment DMN → APPROVE)  
 TRADE\_CARDS: TC-SILVER01 +649 points (loyalty accrual)

### TC-HIRE-01 — hire with FixPro maintenance

RENTALS: BK-... PWH-2500 £67.50, tier "2-3 days" (hire\_pricing DMN → 0.90 multiplier)  
 SERVICE\_RECORDS: SRV-... REPORTED, routine 1 / repair 0 (maintenance\_triage DMN)  
 SERVICE\_INVOICES: INV-... £47.50 (labour £12.50 + call-out £35.00)

## TC-BUY-03 — finance declined (credit score 400)

FINANCE\_TRANSFERS: FIN-REJ-... REJECTED (finance\_assessment DMN → DECLINE)  
No ORDERS row; journey ends at "Customer cancelled".

Final database state after the suite: 3 orders, 3 finance transfers (2 APPROVED / 1 REJECTED), 3 rentals, 2 service invoices, 5 trade cards.

## Operate evidence

| File                                        | Shows                                                                         |
|---------------------------------------------|-------------------------------------------------------------------------------|
| docs/evidence/buy-journey-collaboration.png | Buy-with-finance: Customer + FinTrust pools, 3 call activities, message flows |
| docs/evidence/hire-journey-3pools.png       | Hire-with-maintenance: Customer + FinTrust + FixPro pools                     |
| docs/evidence/fixpro-partner.png            | FixPro partner: collect → triage (DMN) → service → report → return            |
| docs/evidence/dmn-decisions-evaluated.png   | Operate Decisions tab: all four DMN tables evaluated                          |

## Defects / observations

No open defects. Every cancellation path (backorder, quote rejected, finance declined, tool unavailable) reaches a clean *Customer cancelled* end event — the orchestration design has no dead ends.

**Sign-off:** MA — 13 July 2026